Problem · Unit 2 · 10 points · 60 min · Standard exam level

P08 — Seismic accelerometer on a vibrating base: inertia force, frequency response and the two operating regimes

*A vibration sensor is a rigid case whose floor is bolted to a structure that moves **vertically** with the prescribed law $z(t) = Z\cos\nu t$ ($Z > 0$, $\nu > 0$), where z is the height of the case floor above a fixed datum, measured along the upward vertical axis y of the inertial frame S_0. Inside the case, a proof mass m (a particle that can only move along the vertical) is joined to the case floor by a linear spring of stiffness k and natural length L_0 and by a linear viscous damper of constant c (force $-c$ times the velocity of the mass relative to the case). Uniform gravity $-g\,\mathbf{j}$ acts.*

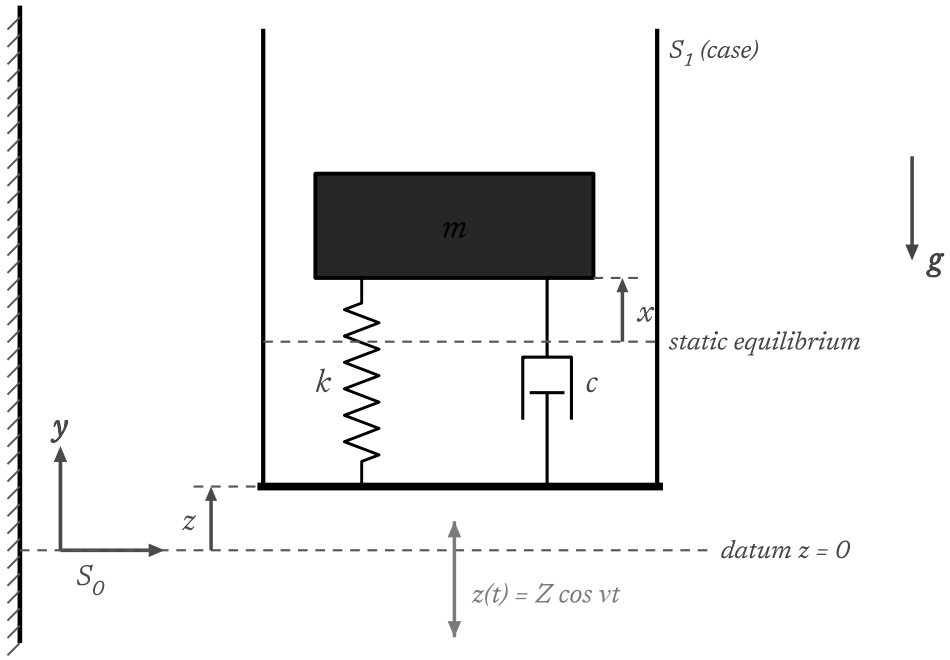


Figure 1. Seismic accelerometer: the case floor moves as $z(t) = Z\cos\nu t$ with respect to the fixed datum of S_0; the proof mass m is held by the spring k and the damper c; x is measured in the case frame S_1 from the static equilibrium level (both z and x positive upwards).

*Let S_1 be the frame attached to the case. The relative coordinate x is the height of the mass in S_1 measured **from its static equilibrium level** (the level where it rests when $z \equiv 0$), positive upwards. Write $\omega_0 = \sqrt{k/m}$ and $\zeta = c/(2\sqrt{km})$ (the Notes' damping parameter is $\gamma = c/(2m) = \zeta\omega_0$), and assume $0 < \zeta < 1$.*

Setup

- Data: m, k, c, L_0, g, Z, ν. Numerical part (f): $m = 0.050$ kg, $k = 500$ N/m, $c = 2.0$ N s/m, $Z = 2.0$ mm, $\nu = 50$ rad/s, $g = 9.81$ m/s^2.
- Hypotheses: point mass with vertical motion only (lateral guides are smooth), massless spring and damper, linear laws, S_0 inertial.
- Frames: $S_0 = \{O; \mathbf{i}, \mathbf{j}\}$ fixed; S_1 attached to the case, a **translating** frame: $\boldsymbol{\omega}_{10} = \mathbf{0}$, origin acceleration $\ddot{z}\,\mathbf{j}$.
- Notation: x is the **vertical** relative coordinate of the mass (along $\mathbf{j}$), not a coordinate along the horizontal axis $\mathbf{i}$ of S_0.

Part (a) · 1.5 points

Degrees of freedom, frames and kinematics

How many degrees of freedom does the mass have? Write the absolute position, velocity and acceleration of the mass in terms of z and x, and identify the relative and the transport (dragging) acceleration.

Step 1. Degrees of freedom

The mass can only move along the vertical: 1 coordinate. The case is not a body of the problem but a prescribed motion, so it adds no unknown; the constraint "the mass moves on the vertical line of the case" is rheonomic in S_0 and fixed in S_1. The coordinate is x.

Step 2. Absolute kinematics

Let ξ be the height of the mass above the case floor, so the absolute height is $y_P = z + \xi$, and $\xi = \xi_e + x$ with ξ_e constant (found in (b)). Differentiating twice (ξ_e is constant):

$$y_P = z + \xi_e + x, \qquad \dot{y}_P = \dot{z} + \dot{x}, \qquad \ddot{y}_P = \ddot{z} + \ddot{x}$$

Step 3. Relative-motion reading

S_1 translates, so $\mathbf{a}_P = \mathbf{a}_{O_1} + \mathbf{a}_{rel}$ (Notes §3.2 with $\boldsymbol{\omega} = \mathbf{0}$): transport acceleration $\ddot{z}\,\mathbf{j}$ with, by the chain rule on $\cos(\nu t)$, $\dot{z} = -\nu Z \sin \nu t$ and $\ddot{z} = -\nu^2 Z \cos \nu t$; relative acceleration $\ddot{x}\,\mathbf{j}$; the Coriolis term $2\boldsymbol{\omega} \times \mathbf{v}_{rel}$ vanishes.

KEY RESULTS

- 1 degree of freedom, $x(t)$; $z(t)$ is prescribed (rheonomic constraint).
- $y_P = z + \xi_e + x$, $\mathbf{a}_P = (\ddot{z} + \ddot{x})\,\mathbf{j}$; $\mathbf{a}_{rel} = \ddot{x}\,\mathbf{j}$, transport $\ddot{z}\,\mathbf{j} = -\nu^2 Z \cos \nu t\,\mathbf{j}$, no Coriolis term.

Part (b) · 1.5 points

Forces in the case frame and static equilibrium

List the forces on the mass in S_1, including the inertia force, in terms of ξ and $\dot\xi$. Find the static equilibrium height ξ_e for $z \equiv 0$ and the static deflection of the spring.

Step 1. Applied forces

- Spring: elongation $\xi - L_0$; the force on the mass is $\mathbf{F}_s = -k(\xi - L_0)\,\mathbf{j}$.
- Damper: opposes the velocity relative to the case, $\mathbf{F}_d = -c\dot\xi\,\mathbf{j} = -c\dot x\,\mathbf{j}$.
- Weight: $\mathbf{W} = -mg\,\mathbf{j}$.
- Guides: a smooth lateral reaction $\mathbf{N}$, horizontal; it has no vertical component and does no work.

Step 2. Inertia force

In the translating frame S_1 the relative-motion law is $m\mathbf{a}_{rel} = \sum \mathbf{F} - m\mathbf{a}_{O_1}$ (Notes §4.9), so

$$\mathbf{F}_{in} = -m\ddot z\,\mathbf{j} = m\nu^2 Z\cos\nu t\,\mathbf{j}$$

When the case accelerates upwards ($\ddot z > 0$) the mass feels pushed down, as in a lift.

Step 3. Static equilibrium

With $z \equiv 0$ there is no inertia force and at rest $\dot\xi = 0$:

$$-k(\xi_e - L_0) - mg = 0 \;\Longrightarrow\; \xi_e = L_0 - \frac{mg}{k}$$

The spring is compressed by $\delta_{st} = mg/k$ and carries the weight.

KEY RESULTS

- $\mathbf{F}_s = -k(\xi - L_0)\,\mathbf{j}$, $\mathbf{F}_d = -c\dot\xi\,\mathbf{j}$, $\mathbf{W} = -mg\,\mathbf{j}$, $\mathbf{F}_{in} = -m\ddot z\,\mathbf{j} = m\nu^2 Z\cos\nu t\,\mathbf{j}$; lateral reaction $\mathbf{N} \perp \mathbf{j}$.
- $\xi_e = L_0 - mg/k$; static deflection $\delta_{st} = mg/k$ (spring compressed).

Part (c) · 1.5 points

Equation of motion

Project the relative-motion law on **j** *and obtain the equation of motion for x. Explain why g does not appear, and write the equation in the normalised form with ω_0 and ζ.*

Step 1. Projection

$$m\ddot{\xi} = -k(\xi - L_0) - c\dot{\xi} - mg - m\ddot{z}$$

Step 2. Change of variable

With $\xi = \xi_e + x, \dot{\xi} = \dot{x}, \ddot{\xi} = \ddot{x}$ and $-k(\xi_e + x - L_0) = -kx - k(\xi_e - L_0) = -kx + mg$ (from (b)):

$$m\ddot{x} = -kx + mg - c\dot{x} - mg - m\ddot{z} \implies m\ddot{x} + c\dot{x} + kx = -m\ddot{z} = m\nu^2 Z\cos\nu t$$

The weight is a constant force; measuring x from the static equilibrium absorbs it into the spring preload, so it drops out of the dynamics. This is true only because the spring is linear: the same constant offset works at every x.

Step 3. Normal form

Dividing by m, with $k/m = \omega_0^2$ and $c/m = 2\zeta\omega_0$:

$$\ddot{x} + 2\zeta\omega_0\dot{x} + \omega_0^2 x = \nu^2 Z\cos\nu t$$

This is Notes Eq. 5.32 with the force amplitude $F_1 = m\nu^2 Z$: the base motion acts as a harmonic force whose amplitude grows with ν^2. Extension (Lagrange equations are not in the course notes): cross-check by Lagrange in S_0: $T = \frac{1}{2}m(\dot{z} + \dot{\xi})^2$, $V = mg(z + \xi) + \frac{1}{2}k(\xi - L_0)^2$ and the damper's generalised force $-c\dot{\xi}$ give the same equation.

KEY RESULTS

- $m\ddot{x} + c\dot{x} + kx = m\nu^2 Z\cos\nu t$.
- $\ddot{x} + 2\zeta\omega_0\dot{x} + \omega_0^2 x = \nu^2 Z\cos\nu t$; g is cancelled by the static preload of the spring.

Part (d) · 2 points

Steady-state response

Using the complex representation, find the amplitude X and the phase φ of the steady-state solution $x_p = X\cos(\nu t + \varphi)$, and write X/Z as a function of $r = \nu/\omega_0$ and ζ. What is the complete solution?

Step 1. Complex equation

The real forcing is $\nu^2 Z\cos\nu t = \mathrm{Re}\big(\nu^2 Z e^{i\nu t}\big)$ (Notes §5.2 and §5.4). With $\tilde{x} = Ae^{i\nu t}$: $\dot{\tilde{x}} = i\nu\tilde{x}$, $\ddot{\tilde{x}} = -\nu^2\tilde{x}$ (chain rule on the exponential), and

$$\big(-\nu^2 + 2i\zeta\omega_0\nu + \omega_0^2\big)A = \nu^2 Z \implies A = \frac{\nu^2 Z}{\omega_0^2 - \nu^2 + 2i\zeta\omega_0\nu}$$

Step 2. Amplitude and phase

Writing $A = Xe^{i\varphi}$, $|A|$ is the modulus of the numerator over that of the denominator:

$$X = \frac{\nu^2 Z}{\sqrt{(\omega_0^2 - \nu^2)^2 + 4\zeta^2\omega_0^2\nu^2}}, \qquad \varphi = -\arg\left(\omega_0^2 - \nu^2 + 2i\zeta\omega_0\nu\right)$$

so $\tan\varphi = -2\zeta\omega_0\nu/(\omega_0^2 - \nu^2)$; the imaginary part of the denominator is positive, hence $-\pi < \varphi < 0$ (the mass lags behind the forcing, Notes Eq. 5.42). The real part is $x_p = X\cos(\nu t + \varphi)$.

Step 3. Dimensionless form

Dividing numerator and denominator by ω_0^2 with $r = \nu/\omega_0$:

$$\frac{X}{Z} = \frac{r^2}{\sqrt{(1 - r^2)^2 + (2\zeta r)^2}}, \qquad \tan\varphi = -\frac{2\zeta r}{1 - r^2}$$

Step 4. Complete solution

Add the general solution of the homogeneous equation (Notes Eq. 5.26, underdamped): $x = X\cos(\nu t + \varphi) + a\,e^{-\zeta\omega_0 t}\cos(\omega_0\sqrt{1-\zeta^2}\,t + \beta)$, with a, β fixed by the initial conditions. The second term is the transient.

KEY RESULTS

- $A = \dfrac{\nu^2 Z}{\omega_0^2 - \nu^2 + 2i\zeta\omega_0\nu}$; $X = |A| = \dfrac{\nu^2 Z}{\sqrt{(\omega_0^2-\nu^2)^2 + 4\zeta^2\omega_0^2\nu^2}}$.
- $\tan\varphi = -\dfrac{2\zeta\omega_0\nu}{\omega_0^2-\nu^2}$ with $-\pi < \varphi < 0$; $\dfrac{X}{Z} = \dfrac{r^2}{\sqrt{(1-r^2)^2 + (2\zeta r)^2}}$.
- $x = X\cos(\nu t + \varphi) + a\,e^{-\zeta\omega_0 t}\cos(\omega_0\sqrt{1-\zeta^2}\,t + \beta)$.

Part (e) · 2 points

Operating regimes and the resonant peak

Discuss the limits $\nu \ll \omega_0$ and $\nu \gg \omega_0$: what does x measure in each case? Then find the forcing frequency at which X is maximum and the maximum value of X/Z. Is it at $\nu = \omega_0$, or at $\omega_0\sqrt{1-2\zeta^2}$ as in Notes §5.4.1?

Step 1. Low frequency: accelerometer

For $r \to 0$ the denominator tends to 1 and $\varphi \to 0$: $X \approx r^2 Z = \nu^2 Z/\omega_0^2$ and

$$x \approx \frac{\nu^2 Z\cos\nu t}{\omega_0^2} = -\frac{\ddot{z}}{\omega_0^2}$$

The relative displacement is proportional to the **acceleration** of the base: the spring force kx supplies the force $-m\ddot{z}$ the mass needs to follow the case.

Step 2. High frequency: vibrometer

For $r \to \infty$: $X/Z \to r^2/r^2 = 1$ and $\tan\varphi = 2\zeta r/(r^2-1) \to 0^+$ with φ in the third quadrant, so $\varphi \to -\pi$, so $x \approx Z\cos(\nu t - \pi) = -z$. The absolute height $y_P = z + \xi_e + x \approx \xi_e$ is constant: the mass cannot follow the fast motion and stays still in S_0, so x measures the **displacement** of the base.

Step 3. Maximum of the response

$(X/Z)^2 = f(s) = s^2/D$ with $D = (1-s)^2 + 4\zeta^2 s$ and $D' = -2(1-s) + 4\zeta^2$. By the quotient rule, $f'(s) = 0$ requires $2sD - s^2 D' = 0$, i.e. $2D = sD'$:

$$2(1-2s+s^2) + 8\zeta^2 s = -2s + 2s^2 + 4\zeta^2 s \implies 2 - 2s + 4\zeta^2 s = 0 \implies s = \frac{1}{1-2\zeta^2}$$

which is admissible ($s > 0$) for $\zeta < 1/\sqrt{2}$. Then $1 - s = -2\zeta^2/(1 - 2\zeta^2)$ and

$$D = \frac{4\zeta^4 + 4\zeta^2(1 - 2\zeta^2)}{(1 - 2\zeta^2)^2} = \frac{4\zeta^2(1 - \zeta^2)}{(1 - 2\zeta^2)^2}, \qquad f = \frac{s^2}{D} = \frac{1}{4\zeta^2(1 - \zeta^2)}$$

so $\nu_{peak} = \omega_0/\sqrt{1 - 2\zeta^2}$ and $(X/Z)_{max} = 1/\left(2\zeta\sqrt{1 - \zeta^2}\right)$.

Step 4. The trap

The Notes' resonant frequency $\sqrt{\omega_0^2 - 2\gamma^2} = \omega_0\sqrt{1 - 2\zeta^2}$ (Eq. 5.46) is for a force of **constant** amplitude F_1. Here the forcing $m\nu^2 Z$ grows with ν^2, which shifts the maximum of the relative amplitude **above** ω_0. Neither $\nu = \omega_0$ (where $X/Z = 1/(2\zeta)$) nor $\omega_0\sqrt{1 - 2\zeta^2}$ is the peak.

KEY RESULTS

- $\nu \ll \omega_0$: $x \approx \nu^2 Z\cos\nu t/\omega_0^2 = -\ddot{z}/\omega_0^2$ (accelerometer). $\nu \gg \omega_0$: $X \to Z$, $\varphi \to -\pi$, $x \approx -z$ (the mass stays still in S_0: vibrometer).
- Peak at $r^2 = \dfrac{1}{1 - 2\zeta^2}$, i.e. $\nu_{peak} = \dfrac{\omega_0}{\sqrt{1 - 2\zeta^2}} > \omega_0$ (for $\zeta < 1/\sqrt{2}$); $\left(\dfrac{X}{Z}\right)_{max} = \dfrac{1}{2\zeta\sqrt{1 - \zeta^2}}$.

Part (f) · 1.5 points

Numerical case

With the data of the Setup, compute ω_0, ζ, the static deflection, X, φ, the energy dissipated per cycle, the decay of the transient, and the error of an accelerometer that reads $\omega_0^2 X$ as the base acceleration amplitude. Locate the resonant peak.

Step 1. Parameters

$\omega_0 = \sqrt{500/0.050} = 100$ rad/s, $\zeta = 2.0/(2\sqrt{500 \times 0.050}) = 2.0/10 = 0.2$, $r = 50/100 = 0.5$, $\delta_{st} = mg/k = 0.050 \times 9.81/500 = 9.81 \times 10^{-4}$ m $= 0.981$ mm.

Step 2. Amplitude and phase

$$\frac{X}{Z} = \frac{0.25}{\sqrt{0.75^2 + 0.2^2}} = \frac{0.25}{\sqrt{0.6025}} = 0.3221 \implies X = 0.644 \text{ mm}, \qquad \varphi = -\arctan\frac{0.2}{0.75} = -14.93^\circ$$

Step 3. Energy per cycle

With $\dot{x} = -\nu X \sin(\nu t + \varphi)$ and $\int_0^{2\pi/\nu} \sin^2(\nu t + \varphi)\, dt = \pi/\nu$ (the antiderivative of $\sin^2 u$ is $\frac{u}{2} - \frac{\sin 2u}{4}$, evaluated between $u = \varphi$ and $u = \varphi + 2\pi$):

$$W = c\nu^2 X^2 \frac{\pi}{\nu} = \pi c \nu X^2 = \pi \times 2.0 \times 50 \times (6.442 \times 10^{-4})^2 = 1.30 \times 10^{-4}\ \text{J}$$

In steady state this equals the work done per cycle by the inertia force $m\nu^2 Z \cos \nu t$ on the relative motion.

Step 4. Transient

The homogeneous part decays as $e^{-\zeta\omega_0 t} = e^{-20t}$: relaxation time $1/(\zeta\omega_0) = 0.05$ s, and $e^{-20t} = 0.01$ at $t = \ln 100/20 = 0.230$ s. It oscillates at $\omega_d = \omega_0\sqrt{1-\zeta^2} = 97.98$ rad/s.

Step 5. Accelerometer error and peak

The true acceleration amplitude is $\nu^2 Z = 2500 \times 0.002 = 5.0$ m/s^2; the instrument reads $\omega_0^2 X = 10^4 \times 6.442 \times 10^{-4} = 6.44$ m/s^2, i.e. $+28.8$ %: at $r = 0.5$ with $\zeta = 0.2$ the sensor is too close to resonance. The peak is at $\nu_{peak} = 100/\sqrt{0.92} = 104.3$ rad/s with $X/Z = 1/(0.4\sqrt{0.96}) = 2.552$, $X = 5.10$ mm.

KEY RESULTS

- $\omega_0 = 100$ rad/s, $\zeta = 0.2$, $r = 0.5$, $\delta_{st} = 0.981$ mm.
- $X = 0.644$ mm, $\varphi = -14.93°$; $W = \pi c\nu X^2 = 1.30 \times 10^{-4}$ J per cycle.
- Transient $\propto e^{-20t}$ (relaxation time 0.05 s, below 1 % after 0.230 s), $\omega_d = 97.98$ rad/s.
- $\omega_0^2 X = 6.44$ m/s^2 against $\nu^2 Z = 5.0$ m/s^2: $+28.8$ % error; peak at $\nu = 104.3$ rad/s with $X = 2.552\ Z = 5.10$ mm.

Interpretation and checks

- **Dimensional check:** $\omega_0^2 = k/m$ in (N/m)/kg = s^{-2}; $\zeta = c/(2\sqrt{km})$ in (N s/m)/(kg s^{-1}) = 1; $\nu^2 Z$ in m s^{-2}, the same as $\omega_0^2 x$; $W = \pi c\nu X^2$ in (N s/m)(s^{-1})(m^2) = J.
- **Limits:** $\zeta \to 0$ gives $X/Z = r^2/|1-r^2|$, infinite at $r = 1$; $Z \to 0$ leaves the free damped oscillator of Notes §5.3. With $\zeta \approx 0.7$ the factor $1/\sqrt{(1-r^2)^2 + (2\zeta r)^2}$ stays close to 1 over a wide band (at $r = 0.5$ it is 0.975, a -2.5 % error), which is why real accelerometers are built with that damping.
- **Common mistakes in the exam:** (1) writing the inertia force as $+m\ddot{z}$, or forgetting it and writing the spring force with the absolute coordinate; (2) keeping mg in the equation after measuring x from the static equilibrium (or dropping it when x is measured from the natural length); (3) taking the peak at $\nu = \omega_0$ or at $\omega_0\sqrt{1-2\zeta^2}$; (4) choosing φ in the wrong quadrant with the arctangent above resonance.